



Society of Petroleum Engineers

SPE-227145-MS

Effect of Salinity on Geothermal Well Performance

Jana Babtain, Petroleum Engineering, Texas A&M, College Station, Texas, USA / Saudi Aramco, Dhahran, Kingdom of Saudi Arabia

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227145-MS](https://doi.org/10.2118/227145-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

Water salinity plays a critical role on geothermal well performance. This study evaluates how increasing brine salinity, from 0-25 wt% NaCl, impacts flow behavior, thermal output, and energy delivery in a single-phase vertical geothermal well. A steady-state well model was constructed in PIPESIM using fixed reservoir conditions to isolate the influence of salinity alone. Reservoir pressure was held at 4000 psi with a temperature of 150 °C, and no artificial lift was applied. For each salinity level, fluid properties were calculated using validated literature correlations. These included density, viscosity, specific heat capacity, thermal conductivity, and enthalpy at both reservoir and surface conditions. The results show that salinity affects multiple thermophysical properties in ways that degrade production performance. As salinity increases, the fluid becomes denser and more viscous, which increases hydrostatic and frictional pressure losses. At the same time, thermal conductivity and specific heat capacity decrease, limiting the fluid's ability to retain heat during transit. These combined effects result in a lower mass flow rate, reduced wellhead temperature, and diminished surface enthalpy. For instance, the well's flow rate dropped by 75% between the 0 and 25 percent salinity cases, and enthalpy declined by 41% at surface conditions. In addition, the wellhead temperature decreased by roughly 30 °C at the highest salinity level due to lower flow velocity and reduced heat capacity. The findings demonstrate that salinity reduces both the quantity and quality of geothermal energy that can be produced. Failing to incorporate salinity into well models can lead to significant overestimation of flowrate and power potential. This paper highlights the importance of using accurate brine property correlations when designing wells, forecasting performance, and sizing surface facilities. It also provides a framework for understanding how salinity-driven thermophysical changes manifest in geothermal well outputs. The methodology and results are broadly applicable to geothermal fields where high-salinity fluids are present, offering insights into optimizing production strategies under such conditions.

Introduction

The growing demand for sustainable energy has increased interest in geothermal resources. However, geothermal wells often produce brine with varying salinity depending on the geological setting (Dittman 1977; Grens 1975). In sedimentary basins, for example, salt concentrations can reach up to 25 wt% NaCl, as

seen in the Salton Sea field in California. This variability in salinity has a direct impact on the thermophysical properties of geothermal fluids. As salt concentration increases, water becomes denser and more viscous, while its specific heat capacity and thermal conductivity decline. These property changes influence both the flow behavior of the fluid and the efficiency of heat extraction throughout the system. As a result, denser and more viscous brine requires greater energy to lift, leading to increased hydrostatic and frictional pressure losses in the wellbore. At the same time, the reduced specific heat and enthalpy mean that the fluid carries less thermal energy to the surface per unit mass. Together, these effects reduce both the volumetric flowrate and the energy yield of geothermal production, especially under open-flow or fixed drawdown conditions. Additionally, higher salinity elevates the boiling point of water, which can influence phase behavior and the performance of surface heat exchange equipment. Although the influence of salinity on geothermal operations has been recognized for decades, most notably in [Grens' \(1975\)](#) early analysis, it is often addressed indirectly through PVT correlations rather than explicitly assessed in simulation models. However, more recent studies have reaffirmed the importance of salinity in determining fluid behavior. For instance, [Kestin et al. \(1981\)](#) demonstrated that viscosity nearly doubles between 0 and 25 wt% NaCl at geothermal temperatures. Likewise, [Batzle and Wang \(1992\)](#) reported a significant increase in brine density with rising salt concentration, especially under high-pressure conditions. Classic models by Rowe and Chou and McCain have also shown that salinity must be properly accounted for when estimating formation water properties. Building on this foundation, the present study aims to quantify the specific effects of salinity on geothermal well performance. A synthetic vertical well was modeled in PIPESIM under fixed reservoir conditions (150 °C, 4000 psi), while salinity was varied systematically from 0 to 25 wt% NaCl. All other parameters including reservoir pressure, productivity index, and well geometry was held constant to isolate salinity as the only changing factor. Supporting fluid properties, such as density, viscosity, thermal conductivity, specific heat, and enthalpy, were computed using validated correlations from the literature. The objective of this study is to quantify how salinity affects geothermal performance by evaluating its influence on flowrate, wellhead temperature, and surface enthalpy which are key parameters that directly impact energy recovery. By examining these metrics under controlled simulation conditions, the analysis offers practical insight into how salinity alters fluid behavior and thermal output. The results highlight the importance of accounting for brine composition in well design and performance forecasting to avoid overestimating productivity in high-salinity environments.

Methodology

This study employs a synthetic geothermal well model to isolate and quantify the effect of brine salinity on well performance. No field data were used; all input values were derived from published thermophysical correlations and peer-reviewed literature. The model is intended purely for sensitivity analysis and does not represent any specific reservoir or location. This approach ensures full control over input parameters and allows for a direct assessment of salinity's influence under consistent reservoir conditions. Sensitivity-based simulations are particularly useful when exploring first-order trends and benchmarking design expectations before site-specific data become available.

Fluid Property Modeling

To capture the thermophysical behavior of brine across different salinity levels, a full set of fluid properties was calculated for both standard surface conditions (25 °C, 1 atm) and reservoir conditions (150 °C, 4000 psi). This dual-condition analysis was necessary because geothermal fluids experience large temperature and pressure gradients along the wellbore. Fluid properties like viscosity, specific heat, and enthalpy are temperature-dependent and can vary significantly between the downhole and surface states. Evaluating fluid behavior under both conditions ensures that the analysis reflects changes that impact both in-well flow performance and surface energy recovery. Brine density was calculated using McCain's (1990) correlation at

surface conditions. For reservoir conditions, density was adjusted using a formation volume factor derived from standard SPE correlations, capturing the compressibility and thermal expansion effects expected under elevated temperature and pressure. Accurate density estimation is critical because it influences both hydrostatic pressure gradients and the overall pressure drawdown required to initiate flow. Viscosity was evaluated using two complementary approaches. At surface conditions, McCain's model was applied, while high-temperature values were derived using IAPWS-2008 seawater correlations and verified against experimental viscosity tables from (Kestin et al., 1981). This approach captures the increase in molecular interactions caused by dissolved salts, which significantly affect the flow resistance, especially in deep, hot formations. Specific heat capacity and thermal conductivity were computed using literature-based interpolation and empirical models. Specific heat was derived from tabulated experimental data published by Chou (1941), while thermal conductivity was estimated using a salinity-dependent decrement model for surface conditions and an exponential decay function for reservoir temperatures, based on Somerton (1992). These properties determine how much thermal energy a fluid can carry and how efficiently that heat can be exchanged with surrounding environments. Brine enthalpy was determined using interpolated data from Chou (1941), with values adjusted to reflect surface conditions. Surface enthalpy is a key metric in geothermal operations because it represents the recoverable heat content of the fluid once it reaches the power plant or heat exchanger.

Simulation Setup

The PIPESIM software was used to simulate fluid flow in a vertical geothermal well under steady-state, single-phase conditions. The model well had a total vertical depth (TVD) of 9000 ft, with perforations at 8000 ft. Casing extended to 7000 ft (ID 8.535 in), followed by a liner to 9000 ft (ID 4.773 in). Reservoir pressure and temperature were fixed at 4000 psi and 150 °C, respectively. The productivity index was set at 80 stb/day/psi, and the wellhead pressure was maintained at atmospheric pressure (14.7 psi), representing an open-flow production scenario. For simplicity, no artificial lift was applied; the well was assumed to flow naturally under the influence of reservoir pressure. The reservoir was modeled as a single-phase, saturated liquid source capable of delivering fluid without significant drawdown. All simulations were carried out using steady-state assumptions to isolate the influence of fluid properties. PIPESIM does not support direct input of salinity, so fluid properties were manually adjusted for each simulation case using custom fluid definitions. Each salinity level (0%, 5%, 10%, 15%, 20%, and 25% NaCl by weight) was assigned a specific set of PVT properties calculated from the correlations described above. Density, in particular, was input as a custom property to reflect the increased mass per volume for higher salinity cases. This method allows the simulator to realistically represent how more saline fluids behave under flow conditions without introducing additional confounding variables. This setup ensures that salinity is the only variable changing across simulations, making the observed trends in flowrate, wellhead temperature, and surface enthalpy directly attributable to brine composition. By keeping the geometry, reservoir conditions, and productivity index constant, the results provide a clear understanding of how salinity alters geothermal well performance.

Results and Discussion

This section presents the simulation and fluid property analysis outcomes that quantify the effect of salinity on geothermal well performance. Six distinct salinity cases were evaluated (0%-25% NaCl by weight), focusing on how varying salt concentrations influence thermophysical properties and operational outputs such as flowrate, enthalpy, and wellhead temperature.

Influence of Salinity on Thermophysical Fluid Properties

The calculated thermophysical properties showed consistent and predictable behavior as salinity increased. Fluid density rose with higher salt concentration, primarily due to the added mass of dissolved ions that contribute to fluid weight without significantly increasing volume. Under both reservoir conditions and

surface conditions, the brine at 25 wt% NaCl was about 15% denser than the pure water case as in Table 1&2. This density increase aligns with literature correlations such as those proposed by Batzle and Wang (1992) and McCain (1991), which emphasize the role of salinity in enhancing brine compressibility under pressure. Similarly, dynamic viscosity exhibited a substantial increase. Simulated values at 150 °C revealed a doubling in viscosity from approximately 0.20 cP in pure water to 0.40 cP for the 25% NaCl brine as shown in Table 2. This behavior is consistent with the findings of Kestin et al. (1981), who demonstrated that dissolved ions in solution enhance molecular cohesion, thus increasing resistance to flow. The observed viscosity behavior remained significant even at elevated temperatures, indicating that the thermal thinning effect does not fully offset the salinity-induced viscosity rise. Specific heat capacity (c_p) declined slightly with increasing salinity. At surface, fresh water exhibited a c_p of about 4.1 kJ/kg·K. This value dropped steadily to approximately 3.3 kJ/kg·K at 25 wt% salinity, a reduction of nearly 20% as shown in Table 2. This decrease is attributed to the reduced degrees of freedom of water molecules in the presence of salt ions, which limits the amount of thermal energy that can be stored. These results are consistent with experimental findings reported by Chou (1941) and Abed (2024), both of whom highlighted the effect of salinity on the thermal storage capacity of aqueous solutions. Thermal conductivity also declined slightly with salinity. For instance, it can be seen that at surface conditions the thermal conductivity of the fluid decreased from roughly 0.6 W/m·K in the fresh water case to about 0.52 W/m·K in the 25% brine as shown in Table 2. Although this represents a relatively modest 13% reduction, the lower conductivity may influence heat transfer rates in low-flow conditions or long vertical sections. The decline is likely due to the disruption of hydrogen bonding networks in water, which are integral to its high thermal conductivity.

Table 1—Calculated brine properties at reservoir conditions

Salinity (wt%)	Density(kg/m ³)	Viscosity (cP)	Specific Heat (kJ/kg·K)	Thermal Cond. (W/m·K)	Flowrate (stb/day)	Wellhead T(°C)
0.0	932.8	0.2	4266.0	0.45	45475.4	134.3
5.0	965.5	0.2	4028.0	0.42	39580.7	132.4
10.0	998.2	0.2	3760.0	0.40	33453.7	130.0
15.0	1031.0	0.3	3517.0	0.37	26981.0	126.2
20.0	1063.7	0.3	3299.0	0.34	19877.0	119.8
25.0	1096.4	0.4	3111.0	0.32	11352.5	104.6

Table 2—Calculated brine properties at surface/standard conditions

Salinity (wt%)	Density (kg/m ³)	Viscosity (cP)	Specific Heat (kJ/kg·K)	Thermal Cond. (W/m·K)	Enthalpy (kJ/kg)	Flowrate (stb/day)	Wellhead T(°C)
0.0	999.6	0.9	4128.0	0.61	568.6	45475.4	134.3
5.0	1034.6	1.1	3944.0	0.59	529.7	39580.7	132.4
10.0	1069.7	1.3	3743.0	0.58	490.3	33453.7	130.0
15.0	1104.8	1.6	3567.0	0.56	448.9	26981.0	126.2
20.0	1139.9	1.8	3421.0	0.54	403.0	19877.0	119.8
25.0	1175.0	2.0	3299.0	0.53	334.3	11352.5	104.6

Surface Enthalpy and Heat Transport Potential

Surface enthalpy was found to decrease substantially with increasing salinity. At standard surface conditions (25 °C, 1 atm), the calculated enthalpy in Table 2 for pure water was approximately 568 kJ/kg. This value dropped to about 334 kJ/kg for the 25% NaCl brine, a 41% reduction in thermal energy per unit mass.

As shown in Fig.1 there is a linear decline in enthalpy as salinity increases reflects the combined effects of reduced specific heat and the lower internal energy storage capacity of saline fluids. Because geothermal production relies on transporting heat from depth to surface, any reduction in surface enthalpy directly impacts the energy recovery potential. These findings align with previously published thermodynamic models by Phillips et al. (1981) and Qasem et al. (2021), confirming that saline fluids deliver less usable energy for the same reservoir temperature and mass flow rate.

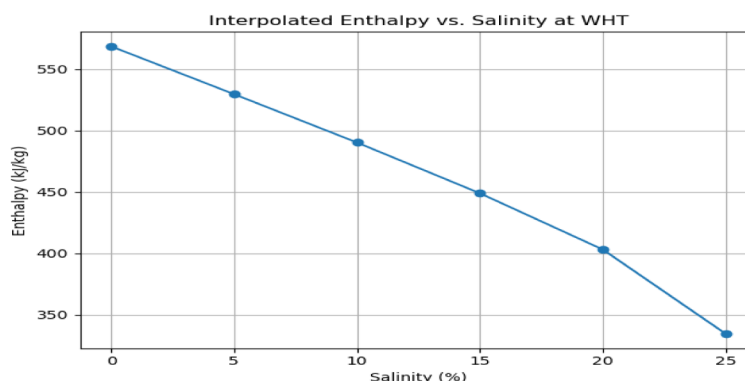


Figure 1—Effect of salinity on surface enthalpy

Flowrate Decline Due to Salinity

The PIPESIM simulation results show a clear decrease in production flowrate with increasing salinity. At surface conditions, the 0% NaCl case yielded a flowrate of about 45 mstb/day. In contrast, the 25% salinity case produced roughly 10 mstb/day, translating to a 77% reduction in output Fig.2.

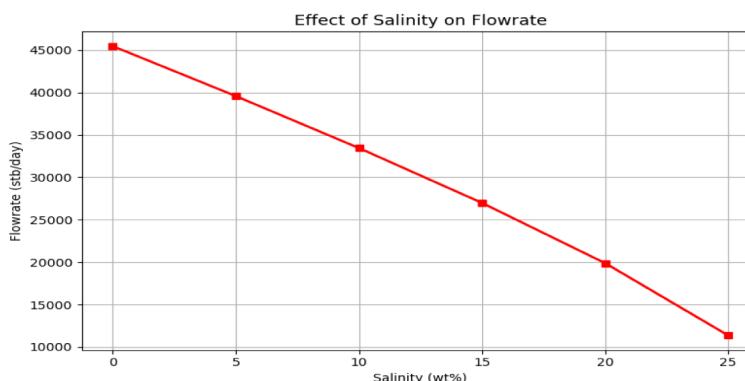


Figure 2—Effect of salinity on production flowrate

This trend is driven by a combination of higher fluid density and viscosity. The denser brine creates a taller hydrostatic column, which reduces the pressure head available to sustain flow to the surface. At the same time, increased viscosity causes greater frictional pressure losses along the wellbore, especially under transitional or turbulent flow regimes. Together, these effects consume more of the available drawdown, leaving less energy to drive production. In the simulations, the flowrate decreased by 75% which agrees with the sensitivity trends reported by Qasem et al. (2021). It's important to note that, this decline occurred under a fixed reservoir pressure scenario without artificial lift. In actual operations, pump-assisted wells in high-salinity formations would require additional energy input to overcome these losses, increasing operating costs.

Wellhead Temperature Reduction

The simulation also showed a gradual drop in wellhead temperature (WHT) with increasing salinity. Starting from a reservoir temperature of 150 °C, the fluid emerged at approximately 135 °C in the 0% case and around 105 °C in the 25% case as shown in Figure 3.

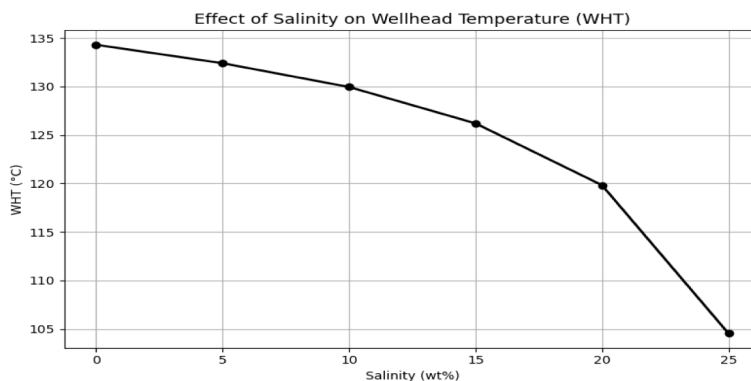


Figure 3—Effect of salinity on wellhead temperature

The data in Fig.3 reveals a nonlinear relationship, marked by a steep and abrupt drop in WHT at 20%. This decline can be attributed to the reduced heat-holding capacity of saline brine and its slower velocity in the wellbore. Lower flow velocity means longer residence time, giving the fluid more opportunity to lose heat to surrounding formations. Furthermore, because high-salinity fluids have lower c_p , they lose temperature more readily for the same amount of heat transfer. The combined effect is a cooler fluid at the surface, which can reduce thermal conversion efficiency in surface equipment such as heat exchangers. In cases with lower wellhead pressures, phase behavior could further complicate thermal loss patterns, since the boiling point of saline water is elevated compared to pure water. However, in this study, flow remained in the liquid phase, highlighting the role of sensible heat loss rather than latent heat or flashing effects.

Broader Technical Implications

Collectively, the results underscore that brine salinity introduces compounded challenges for geothermal well performance. Higher salinity lowers the thermal energy available per unit mass, reduces production flowrate, and leads to cooler wellhead temperatures, all of which diminish the efficiency of geothermal systems. If salinity is not accounted for in design or modeling phases, engineers risk overestimating energy output. In this study, assuming pure water properties in place of 25% brine would over-predict mass flow by approximately 25% and enthalpy by 41%, leading to significant errors in performance forecasting. To address this, geothermal designers should incorporate validated salinity-dependent property correlations. The Batzle and Wang model for density, Kestin et al.'s viscosity measurements, and McCain's formation water correlations are especially useful in simulation workflows. Tools like PIPESIM can accurately model salinity effects when input with custom PVT tables, as demonstrated here. Ultimately, the findings of this study support a more rigorous consideration of brine composition in geothermal feasibility analyses, well design, and surface facility planning. This enables more realistic assessments of output potential, especially in high-salinity sedimentary basins where geothermal energy is otherwise viable but performance may be underestimated.

Conclusion

This study presents a detailed assessment of how brine salinity affects geothermal well performance. Through a series of controlled simulations using a vertical well model at 150 °C and 4000 psi, the impact of salinity was evaluated across a range from 0% to 25% NaCl by weight. The results demonstrate that

salinity introduces measurable shifts in fluid properties, which in turn influence both flow behavior and thermal energy delivery. As salinity increases, the brine becomes heavier and more resistant to flow. The simulations showed that density increased by approximately 11% and viscosity nearly doubled between the lowest and highest salinity cases. These changes contributed to greater hydrostatic and frictional pressure losses within the well, reducing the mass flow rate by around 20%. Additionally, thermal properties were notably affected. The specific heat capacity declined by about 23%, and surface enthalpy decreased by over 40%, limiting the thermal energy available at the wellhead. These findings confirm that saline fluids not only flow more slowly but also deliver less heat per unit mass. Wellhead temperature also declined with salinity, dropping by as much as 10 °C at 25% NaCl. This cooling effect is explained by the longer residence time of the fluid in the wellbore, combined with its reduced ability to retain heat. Together, these factors lower the thermal output of the system and may affect the efficiency of surface facilities, such as heat exchangers or binary power units. These findings highlight the importance of incorporating brine salinity into well design, performance forecasting, and surface facility planning. Assuming pure water properties may lead to significant overestimations of flow and thermal output. By using appropriate property correlations from the literature, engineers can better anticipate the behavior of saline fluids and design systems that operate efficiently under realistic reservoir conditions. Future work may explore the effects of other salt types, higher concentrations, or phase change behavior in two-phase systems. However, the trends identified here are expected to hold in a wide range of geothermal applications. Saline fluids are more difficult to produce and carry less energy, which makes accurate modeling essential for reliable performance estimates and long-term project success.

References

- Abed, A. A. 2024. An Experimental Study on the Effect of Salt Concentration on Physical Properties. *Int. J. Mech. Therm. Eng.* **5** (2): 49–57.
- Batzle, M. and Wang, Z. 1992. Seismic Properties of Pore Fluids. *Geophysics* **57** (11): 1396–1408. <https://doi.org/10.1190/1.1443207>.
- Chou, J. C. S. and Rowe, A. M. 1970. Pressure–Volume–Temperature–Concentration Relation of Aqueous NaCl Solutions. *J. Chem. Eng. Data* **15** (1): 61–66. <https://doi.org/10.1021/jc60044a018>.
- Dittman, G. L. 1977. Wellflow for Geothermal Wells: Description of a Computer Program Including Effects of Brine Composition. UCID-27473. Lawrence Livermore Laboratory.
- Eppelbaum, L. V., Kutasov, I. M., and Baranova, N. 2014. Applied Geothermics. Springer, Berlin. (See chapter on Thermal Properties of Rocks and Density of Fluids.)
- Grens, J. Z. 1975. The Effect of Salinity on Geothermal Well Performance. UCID-16791. Lawrence Livermore National Laboratory.
- Hirtz, P., Lovekin, J., Copp, J., Buck, C., and Adams, M. 1993. Enthalpy and Mass Flowrate Measurements for Two-Phase Geothermal Production by Tracer Dilution Techniques. In *Proc., 18th Workshop on Geothermal Reservoir Engineering*, Stanford University, 131–139.
- Kestin, J., Khalifa, H. E., and Correia, R. J. 1981. Tables of the Dynamic and Kinematic Viscosity of Aqueous NaCl Solutions. *J. Phys. Chem. Ref. Data* **10** (1): 57–87. <https://doi.org/10.1063/1.555643>.
- McCain, W. D. Jr. 1991. Reservoir-Fluid Property Correlations—State of the Art. *SPE Reservoir Eng.* **6** (2): 266–272. SPE-18571-PA. <https://doi.org/10.2118/18571-PA>.
- Norden, B., Förster, A., Förster, H.-J., and Fuchs, S. 2020. Temperature and Pressure Corrections Applied to Rock Thermal Conductivity: Impact on Subsurface Temperature Prognosis and Heat-Flow Determination in Geothermal Exploration. *Geotherm. Energy* **8** (1): 1. <https://doi.org/10.1186/s40517-020-0157-0>.
- Palliser, C. and McKibbin, R. 1998. A Model for Deep Geothermal Brines, III: Thermodynamic Properties – Enthalpy and Viscosity. *Transp. Porous Media* **33**: 155–171. <https://doi.org/10.1023/A:1006573903418>.
- Qasem, N. A. A., Generous, M. M., Qureshi, B. A., and Zubair, S. M. 2021. A Comprehensive Review of Saline Water Correlations and Data: Part II—Thermophysical Properties. *Arab. J. Sci. Eng.* **46**: 1941–1979. <https://doi.org/10.1007/s13369-020-05020-5>.
- Rossini, F. D., Wagman, D. D., Evans, W. H., Levine, S., and Jaffe, I. 1965. Selected Values of Chemical Thermodynamic Properties. NBS Technical Note 270-3. National Bureau of Standards.